

Stripe Payment Processing API Documentation

Version 2.0.0

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Overview

The Stripe Payment Processing API provides a comprehensive solution for handling online payments through Stripe's secure checkout system. The API consists of two main endpoints:

1. **Create Stripe Checkout Session** - Initiates payment processing with automatic fee calculation and merchant validation
2. **Check Application Status** - Verifies payment completion status for applications

The API automatically calculates processing fees, validates merchant credentials, and provides secure redirect URLs for payment completion. All transactions are processed through Stripe's PCI-compliant infrastructure.

Supported Features

- **Currency:** USD only (additional currencies coming soon)
- **Payment Methods:** All Stripe-supported payment methods
- **Minimum Transaction:** \$1.00 USD
- **Maximum Transaction:** No limit (subject to merchant account limits)

UAT System Availability

Important System Access Schedule:

- **Weekdays (Mon-Fri):** 9:00 AM - 5:00 PM Bhutan Time (BT)
- **Weekends (Sat-Sun/Govt Holidays):** System is NOT available
- **Beyond working hours testing:** Requires advance internal approval



Base URL(UAT)

<https://internal-gateway.sit.digitalkidu.bt:8082/uat/stripe/>

Authentication

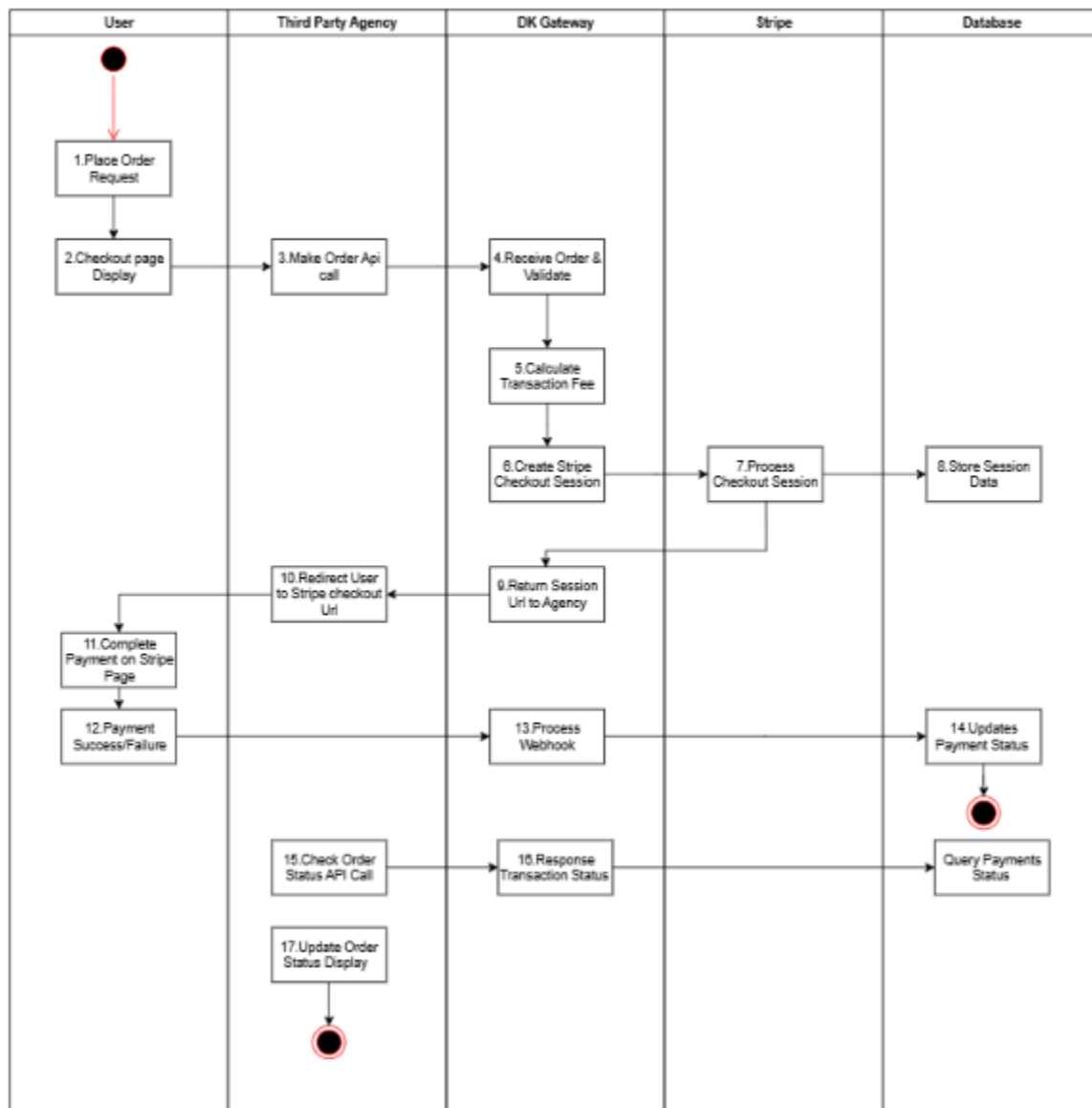
All API requests require authentication using the following header:

Header	Value
X-Gravitee-API-Key	69716f2b-9148-4065-a6c8-934e32f57456

Example Header

X-Gravitee-API-Key: 69716f2b-9148-4065-a6c8-934e32f57456
Content-Type: application/json

Process flow





API Endpoints

Create Checkout Session

Creates a secure Stripe checkout session with automatic fee calculation and merchant validation.

Endpoint Details

- **Method:** POST
- **URL:** /checkout

Request Parameters

Name	Type	Required	Description	Validation
amount	decimal	✓	Base payment amount (minimum 1)	Minimum: 1.00
currency	string	✓	Currency code (only "usd" supported)	Must be "usd"
agency_name	string	✓	Name of the agency processing the payment	Non-empty string
application_no	string	✓	Unique application number for tracking	Must be unique
submerchant_id	string	✓	Sub-merchant identifier for validation	Must exist in system



dk_account	string	✓	DK account identifier for merchant validation	Must exist in system
success_url	string	✓	URL to redirect after successful payment (must be valid HTTPS URL)	Must be valid HTTPS URL
cancel_url	string	✓	URL to redirect if payment is cancelled (must be valid HTTPS URL)	Must be valid HTTPS URL

Sample Request

```
{
  "amount": 100.00,
  "currency": "usd",
  "agency_name": "Digital Services Agency",
  "application_no": "APP20250812001",
  "submerchant_id": "MERCHANT_001",
  "dk_account": "MERCHANT_001",
  "success_url": "https://yourdomain.com/payment/success",
  "cancel_url": "https://yourdomain.com/payment/cancel"
}
```



Success Response (200 OK)

```
{
  "response_code": "0000",
  "response_message": "Success",
  "response_description": "Stripe checkout session created successfully",
  "response_data": {
    "session_id":
"cs_test_a1RWNeorlFfEnmu8lK32XCMzMv9L5R4vzCY5oDeyKJMox86g7T4zbsLT8s",
    "session_url":
"https://checkout.stripe.com/c/pay/cs_test_a1RWNeorlFfEnmu8lK32XCMzMv9L5R4vzCY5oDeyKJMox86g7T4zbsLT8s"
  }
}
```

Response Parameters

Root-Level Fields

Field	Type	Description
response_code	string	Status code of the API call (0000 for success)
response_message	string	Message summarizing the result
response_description	string	Additional description of the response
response_data	string	Contains session details for successful requests



Checkout Session Object Fields (response_data)

Field	Type	Description
session_id	string	Unique Stripe checkout session identifier
session_url	string	Stripe checkout URL for payment completion

Error Responses

400 Bad Request – Validation errors

```
{
  "response_code": "4001",
  "response_message": "Validation error",
  "response_description": "Invalid input: The payment amount must be at least 1 unit"
}
```

400 Bad Request – Invalid currency

```
{
  "response_code": "4001",
  "response_message": "Validation error",
  "response_description": "Invalid currency: eur. Only 'usd' is supported."
}
```



400 Bad Request – Invalid merchant

```
{  
  "response_code": "4001",  
  "response_message": "Invalid sub merchant ID",  
  "response_description": "sub merchant ID cannot be null or empty"  
}
```

400 Bad Request – Invalid request to Stripe

```
{  
  "response_code": "4002",  
  "response_message": "Invalid request",  
  "response_description": "Invalid request to Stripe: [detailed message from Stripe]"  
}
```

404 Not Found – Merchant not found

```
{  
  "response_code": "4004",  
  "response_message": "Merchant not found",  
  "response_description": "Merchant does not exist in merchant table.  
Please verify your merchant details."  
}
```



409 Conflict – Duplicate request

```
{  
  "response_code": "4003",  
  "response_message": "Duplicate request",  
  "response_description": "Duplicate request detected for application  
number APP12345. Please try again with a different request."  
}
```

429 Too Many Requests

```
{  
  "response_code": "4029",  
  "response_message": "Rate limit exceeded",  
  "response_description": "Too many requests. Please try again later."  
}
```

500 Internal Server Error

```
{  
  "response_code": "5000",  
  "response_message": "Internal server error",  
  "response_description": "Internal server error occurred. Please try again  
later."  
}
```



500 Internal Server Error – Verification service error

```
{  
  "response_code": "5006",  
  "response_message": "Verification service error",  
  "response_description": "An error occurred while verifying merchant details.  
Please try again later."  
}
```

500 Internal Server Error – Merchant service timeout

```
{  
  "response_code": "5003",  
  "response_message": "Service timeout",  
  "response_description": "Merchant verification service is temporarily slow.  
Please try again in a few moments."  
}
```

500 Internal Server Error – Merchant service connection failure

```
{  
  "response_code": "5004",  
  "response_message": "Connection timeout",  
  "response_description": "Unable to connect to merchant verification service.  
Please try again later."  
}
```



500 Internal Server Error – Merchant network error

```
{  
  "response_code": "5005",  
  "response_message": "Network error",  
  "response_description": "Network error occurred while verifying merchant.  
Please try again later."  
}
```

501 Authentication Error

```
{  
  "response_code": "5001",  
  "response_message": "Authentication error",  
  "response_description": "Payment service authentication error"  
}
```

502 Service Unavailable

```
{  
  "response_code": "5002",  
  "response_message": "Service unavailable",  
  "response_description": "Payment service temporarily unavailable. Please try  
again later."  
}
```



Check Application Status

Checks if the payment is successful or not.

Endpoint

- **Method:** GET
- **URL:** /check-application-status

Query Parameters

Name	Type	Required	Description
application_no	string	☒	Application number to check for payment completion



Usage Examples

Check if application payment is completed:

```
/check-application-status?application_no=APP12345
```

Response Parameters

Root-Level Fields

Field	Type	Description
response_code	string	Status code of the API call (0000 for success)
response_message	string	Message summarizing the result
response_description	string	Additional description of the response
response_data	boolean	True if payment completed, false if not completed

Success Response

200 OK – Application Found and Completed

```
{  
  "response_code": "0000",  
  "response_message": "Application Found",  
  "response_description": "Application number exists with checkout session completed status.",  
  "response_data": true  
}
```



200 OK – Application Not Found or Not Completed

```
{  
  "response_code": "4004",  
  "response_message": "Application Not Found",  
  "response_description": "Application number does not exist or checkout  
session is not completed.",  
  "response_data": false  
}
```

Error Responses

400 Bad Request – Missing application number

```
{  
  "response_code": "4000",  
  "response_message": "Invalid Input",  
  "response_description": "Application number is required and cannot be empty."  
}
```

400 Bad Request – Validation error

```
{  
  "response_code": "4001",  
  "response_message": "Validation error",  
  "response_description": "Invalid input: [specific validation error]"  
}
```

500 Internal Server Error

```
{  
  "response_code": "5000",  
  "response_message": "Internal Server Error",  
  "response_description": "An error occurred while checking the application."  
}
```



Important Notes

Payment Success Criteria

Key Requirement

Treat the payment as successful ONLY when response_data is true.

All other responses (including false or error responses) should be considered as payment not completed.

Merchant Usage & Service Policy

Important - Applies to All Merchants Integrated with DK Stripe

- If a merchant's transaction causes, or could potentially cause, harm to the company's reputation, or poses a risk to transactions via Stripe, the company reserves the right to suspend or discontinue services for that merchant.
- All customer refund requests must be directed to DK Bank's operations team for processing.
- Non-compliance with these guidelines may result in temporary suspension of services until resolved, as per company terms and conditions.



Error Code Reference

Response Code	Message	Description
0000	Success	Request completed successfully
4000	Invalid Input	General input validation error
4001	Validation error	Specific validation error with details
4002	Invalid request	Invalid request to Stripe payment service
4003	Duplicate request	Duplicate request detected
4004	Not found	Merchant or application not found
4029	Rate limit exceeded	Too many requests sent
5000	Internal server error	General server error
5001	Authentication error	Payment service authentication failed
5002	Service unavailable	Payment service temporarily unavailable

Transaction Fee Calculation

Fee Structure

The API automatically calculates and applies the following fees to all transactions:



- **Stripe Fee: 4.15% of transaction amount**
- **DK Fee: 0.7% of transaction amount**
- **Fixed Fee: \$0.60 per transaction**

Fee Calculation Process

The API performs the following steps for fee calculation:

- 1. Calculate Total Percentage Fee: (Amount × 4.85%)**
 - **Stripe Fee: 4.15%**
 - **DK Fee: 0.7%**
 - **Combined: 4.85%**
- 2. Add Fixed Fee: + \$0.60**
- 3. Calculate Final Amount: Base Amount + Total Fees**
- 4. Convert to Cents: Final amount converted to cents for Stripe processing**

Rounding: All monetary calculations use ROUND_HALF_UP rounding to 2 decimal places.

Fee Examples

\$100 Transaction Breakdown

- **Base Amount: \$100.00**



- **Fee Calculation:**
 - **Stripe Fee:** $\$100.00 \times 4.15\% = \4.15
 - **DK Fee:** $\$100.00 \times 0.7\% = \0.70
 - **Fixed Fee:** $\$0.60$
 - **Total Fees:** $\$4.15 + \$0.70 + \$0.60 = \5.45
- **Final Amount:** $\$100.00 + \$5.45 = \$105.45$
- **Customer pays:** $\$105.45$

Customer Display

The customer will see the final amount including all fees during the Stripe checkout process.

Notes

- **All fees are automatically calculated by the API**
- **Fees are included in the checkout session amount**
- **The merchant receives the base amount minus processing costs**
- **Duplicate Prevention:** Idempotency keys prevent duplicate charges for the same application
- **Merchant Validation:** Both `submerchant_id` and `dk_account` are validated against the merchant database before processing